Reactionof Zinc and Iodine

This experiment is based on one published by Stephen DeMeo in J. Chem. Educ., 72, 1995, 836.
The experimental outline was written by Gordon Bain and revised by Chad C Wilkinson and Léa Gustin.

Academic Goals

- To practice making and recording observations during an experiment.
- To investigate the relationship between mass and stoichiometry.
- To investigate the reversibility of a chemical reaction.
- To prepare a relevant discussion section of a report.

Background

A reaction between two elements to form a binary compound is one of the simplest chemical transformations. A number of factors come into play when reactants are combined to form a product. The first is the ratio of one element to another in the product (or products) of the reaction. The second is the number of *atoms* of each type in one mole of each reactant. One of the reagents in this synthesis is iodine, which exists as a diatomic molecule, written as I_2. When we balance the equation we expect there will be an even number of iodine atoms in it. Iodine's presence as a diatomic molecule on the reactant side of the equation requires this. We summarize these factors under the heading of *stoichiometry*.

We cannot measure out a known number of atoms, but we can measure a known mass of a material. In this experiment, you will react equal masses of zinc and iodine with each other.

Zinc is a gray metal that is used in the form of small granules in this experiment. Solid iodine is a black, shiny solid, but is not a metal according to chemical classifications (it is held together by Van der Waal's forces, not metallic bonds, and does not conduct electricity in the way a metal does). If solid iodine is heated, it sublimes to give a purple vapor. In aqueous solution, if I^- (iodide) ion is present, an equilibrium exists between I_2(aq), I^-(aq) and I_3^-(aq) (triiodide). This aqueous equilibrium mixture has a dark orange-red color.

By measuring the masses of the reactant left over and the product of the reaction, as well as some consideration of the known behavior of zinc and iodine, you will arrive at some conclusions about the stoichiometry of this reaction.

Note that although acidified water is used in this reaction, the amount of acid present is very small and is used only to suppress an undesirable side reaction. The acid plays no part in the reaction between zinc and iodine.

Preparing for the Experiment

- **Review –** Read the procedure for this experiment as outlined in this manual and be sure you are familiar enough with it to prepare the appropriate entries in your notebook. Consider how you will analyze the data you collect and review the appropriate background material found in this manual to accomplish the task. Use the online content to become familiar with techniques and concepts.
- **Pre-lab Video –** Watch the corresponding video online. It is available on your Chem 103 course website.
- **Pre-lab Quiz –** Take the pre-lab quiz available through the Chem 103 course website.
- **Pre-lab Activity –** Complete the activity that is printed in this manual after this experiment.

- **Notebook** – You should prepare your notebook according to the instructions in the pre-lab video. Refer to the Example Experiment (page xxxiii) for an example of a notebook procedure.

  Recording careful observations will be important to the success of this experiment and will be evaluated as part of your grade. Be sure in the two-column procedure/observation format of the lab notebook you leave sufficient space between procedure entries to record a detailed account of what occurs as you perform the experiment.

Techniques in This Experiment

On the Web	In the Manual (Appendices)
• Pre-lab video • Pre-lab quiz	• Accurate Weighing (A2) • Drying to Constant Weight (A2) • Gravity Filtration (A2) • Types of Observations (A2)

Safety Information

Reagent/Equipment	Information	Precaution/Treatment
Iodine	Will stain skin and clothing. Corrosive to metals.	Wear gloves when handling. Rinse affected skin and clothing with water.
Stirring hot plate	The hot plate surface, and any directly heated items, will be hot enough to burn upon contact.	Place afflicted burn under cold water for 10 min.

Disposal Guidelines

Container	Item	Additional Instructions
Trash Can	Gloves	
	Paper products (paper towels, filter paper, weighing paper, etc.)	
	Parafilm	
Sink	Iodine	Flush well with water.
	Ethanol	Flush well with water.
	Reaction product	Flush well with water.
"Used Zinc Metal" Container	Zinc	No liquids in container.

The above guidelines are for expected quantities used by students. In cases of excessive overuse, alternative disposal protocols may be required. Consult with your lab instructor in such cases.

Experimental Procedure

Reagents and Supplies

Washables	Equipment	Reagents/Consumables
5 mL centrifuge tube	3-prong clamp (small)	Zinc
15 mL beaker	Stirring hot plate	Iodine
Spatula	Battery	Acidified water
Funnel	Battery clip	Ethanol
Evaporating dish	Copper wire x 2	Weighing paper
Watch glass	Tongs	Filter paper
	Parafilm	
	Sandpaper	

A. The Reaction

Your reaction vessel for this experiment is a 5 mL centrifuge tube. Use the weighing by difference technique to obtain a mass of iodine for the reaction:

1. Weigh the clean, dry, empty and capped centrifuge tube.
2. At the fume hood, load the tube with iodine **to the first mark on the conical portion of the tube from the bottom**. A spatula is provided for exclusive use with this iodine. Use only this spatula to transfer iodine, and leave it at the station.
3. Cap the tube and weigh again. The difference in mass between the filled and empty tube is the mass of iodine for the reaction (approximately 0.5 g, ***do not use above 0.7 g***).

Important: Double check the amount of reactant you are using. Clear disrespect of the guidelines above may result in dismissal from the lab session (and therefore a zero for the activity).

Fill the tube with acidified water to the 5 mL line.

Weigh out a quantity of zinc **approximately** equal to your mass of iodine and add it to the tube. Cap the tube tightly and then shake vigorously. If you notice the cap leaking, you will need to seal the cap with a small piece (1 inch square) of parafilm.

With vigorous shaking, the reaction will be over in approximately five minutes. At regular intervals, pause briefly in shaking the reaction tube to note observations. The reaction is complete when you can **no longer observe changes** in the contents of the reaction tube. Check with your lab instructor that your reaction is complete before moving on to the next part.

With this experiment you have observed that changes in color can indicate that a chemical reaction has taken place. Are there other observations that can help with this determination (hint: how did the reaction tube feel as you carried out the reaction?).

B. Isolating the Excess Reagent

Weight the clean, dry, empty evaporating dish. Prepare a filtration and evaporation system as shown in Figure 1. This set-up will allow for both filtering and solution evaporation. Set the hot plate to a surface temperature of 120 °C at first.

Filter the reaction mixture by gravity filtration. Rinse the tube and cap onto the filter with about 2 mL of water, followed by 2 mL of ethanol. Once all of the liquid has passed through the filter, rinse the solid with 2 mL of ethanol.

Once this rinse has passed through the filter, remove the filter paper from the funnel. Invert the cone of filter paper and place it on another circle of filter paper, set on the corner of the hot plate. Your excess reagent should be dry and ready to weigh in approximately 15 min. Transfer the excess reagent to weighing paper before you weigh it. Keep it on the lab bench until the end of the experiment. Based on the mass of excess reagent, predict the mass of product formed (theoretical yield) before you measure the mass of the product.

Figure 1. Set-up for gravity filtration and evaporation.

Turn the hot plate up to 200 ºC. Evaporating the solution to dryness will take longer (about an hour). As the evaporation takes place, a salt plate will form over the remaining solution. This plate will slow evaporation or, if it forms a seal, will build pressure from evaporating solvent resulting in ejection of solid from the evaporating dish.

To prevent this salt plate from interfering with the evaporation process, gently roll the evaporating dish and bring the liquid out from under the salt plate. A white to off-white solid should result when the solvent evaporates. This is the product of the reaction. Dry the sample to constant weight as described in Appendix 2.

When you think the evaporation is complete, take the dish off the hot plate and **allow it to cool down before weighing it** (no hot dish on the balance). Place the dish back on the hot plate for a couple of minutes, repeat the weighing process to confirm the mass of product (actual yield).

The reaction product is *deliquescent*. It can absorb water from the atmosphere to the point of dissolving back into a solution. This process takes time, but occurs more rapidly at cooler temperatures.

C. Decomposition of the Product

Add approximately 5mL of deionized water to your product and swirl to dissolve it. Pour this solution into a 15 mL beaker. Obtain a 9 volt battery, a battery cap with color-coded wires and alligator clips and a pair of copper wires to use as electrodes. **Be careful not to short circuit the battery by allowing the alligator clips or wire electrodes to touch while connected to the battery.** Place the tips of both wire electrodes in the solution and attach the cap to the battery. **Note: Do not place the alligator clips into the solution.**

Allow a current to pass for about 2 minutes, record your observations, and identify the chemicals forming at the electrodes. Remove the battery from the cap and leave the electrodes in the solution for 10 minutes (while you start work on your lab report). Record what happens.

Use sandpaper to clean the electrodes. Then return the battery, clips and electrodes and clean up your glassware. Use tap water for cleaning, but rinse all glassware with deionized water (from a wash bottle) before putting it away. You should **not** dry glassware with a paper towel as this would contaminate it with paper fibers. Glassware will air-dry before you need it again.

Data Analysis

Determine the ratio of zinc atoms to iodine atoms that reacted in part A. Do not round up to the next whole number. Determine the formula of the product using a maximum subscript for Zn to be 5. Example: **raw** Zn:I ratio = 1:1.33 chemical formula = Zn_3I_4

Determine the theoretical yield of zinc iodide based on:

- the amount of limiting reagent consumed
- the law of conservation of mass

Calculate the percent yield based on each theoretical yield.

Group Report

Due before leaving lab (even if not done with the post-lab questions)

The **carbon-copy pages of your lab notebook** should be given to your TA in physical form **prior to leaving the lab area**. **Each student** submits their notebook pages with appropriate data, observations and at least one example of any calculation carried out in lab.

Due 24 hours after the end of your lab session (or before leaving lab if done early)

Upload a pdf of your completed report form using the editable pdf template on Canvas. Note: to paste images into the template, you will need the template open in Adobe Reader (the app, not just in-browser). You may upload a single file for your group, or each member may upload their own scan provided each group member indicates whether the submission is the primary one that should be graded. If additional notebook pages were completed after the end of lab session, upload them on Canvas as well.